

RESERVE NETWORK STUDY

One network across the whole reserve.

We modelled Buffalo Kloof's terrain from satellite elevation data and checked every radio path against it. two ridge high-sites give line of sight across 48% of the Protected Environment, fed from a carrier uplink on Makhanda high ground.

Prepared for

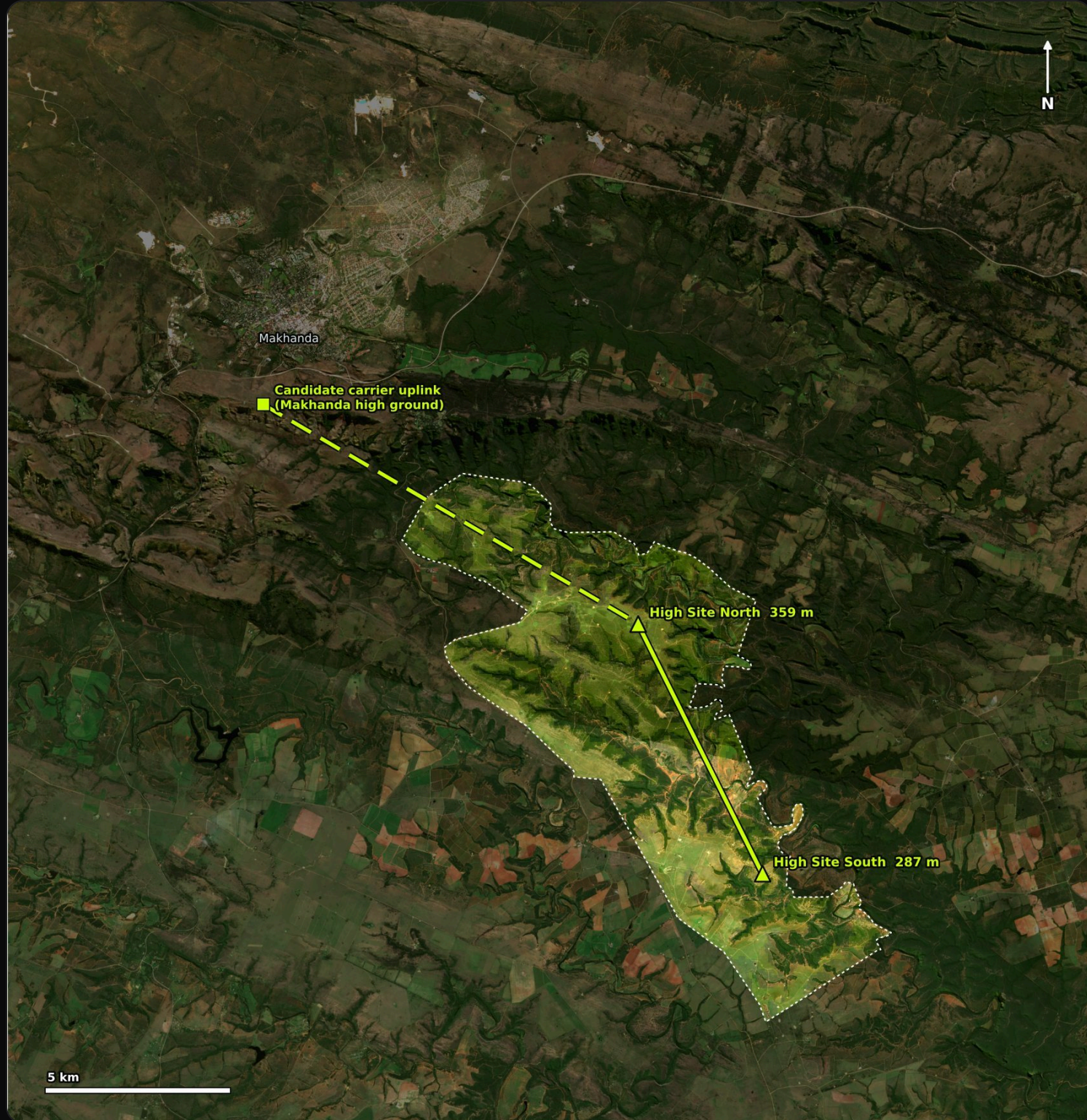
Warne Rippon, Buffalo Kloof

Prepared by

Gerhard Smit, CTTX Services (Pty) Ltd

Date

28 September 2026



--- Carrier uplink
 — Private backbone
 — Lodge link
 Clear line of sight for a 3 m field terminal

Sentinel-2 imagery, 26 March 2026 to 14 July 2026. The dotted outline is the mapped boundary of the Protected Environment, 7,474 ha of the roughly 20,000 ha Buffalo Kloof conservancy. The rest of the conservancy has no public boundary yet and is not modelled. Site positions are desktop candidates, to be confirmed on a site visit.

48%

of the Protected Environment in line of sight

2 + 0

ridge high-sites plus short relays

48%

of the 75 km² study area in line of sight

12.1 km

clear carrier uplink into the reserve

01

Where things stand today

Buffalo Kloof carries black and white rhino, cheetah reintroduction and a dedicated anti-poaching team across the Kowie River valley. The connectivity that is described stops at the camp's living area. We propose one backbone that belongs to the reserve and reaches the field.

WHAT THE PUBLIC RECORD SHOWS

3

02

Carrier in, private network out

Carrier connectivity comes into the reserve once. Everything past that point runs on infrastructure the reserve owns.

LAYER 1 · CARRIER

Uplink from Makhanda high ground

An uncontended business-grade Vodacom service, delivered by CTTX over a 12.1 km link to High Site North. LTE failover keeps the reserve connected if the carrier path drops.

LAYER 2 · BACKBONE

Ridge high-sites

High Site South (287 m) and High Site North (359 m) sit 7.7 km apart with a clear link between them. Each site runs on solar and lithium storage with remote monitoring, so the network stays up through load-shedding.

LAYER 3 · DISTRIBUTION

Every lodge and ops site

Sector radios reach the lodges, HQ, the anti-poaching base, ecology, the workshop and the gates. A LoRaWAN gateway is ready for fence, water-point, pump and fuel sensors.

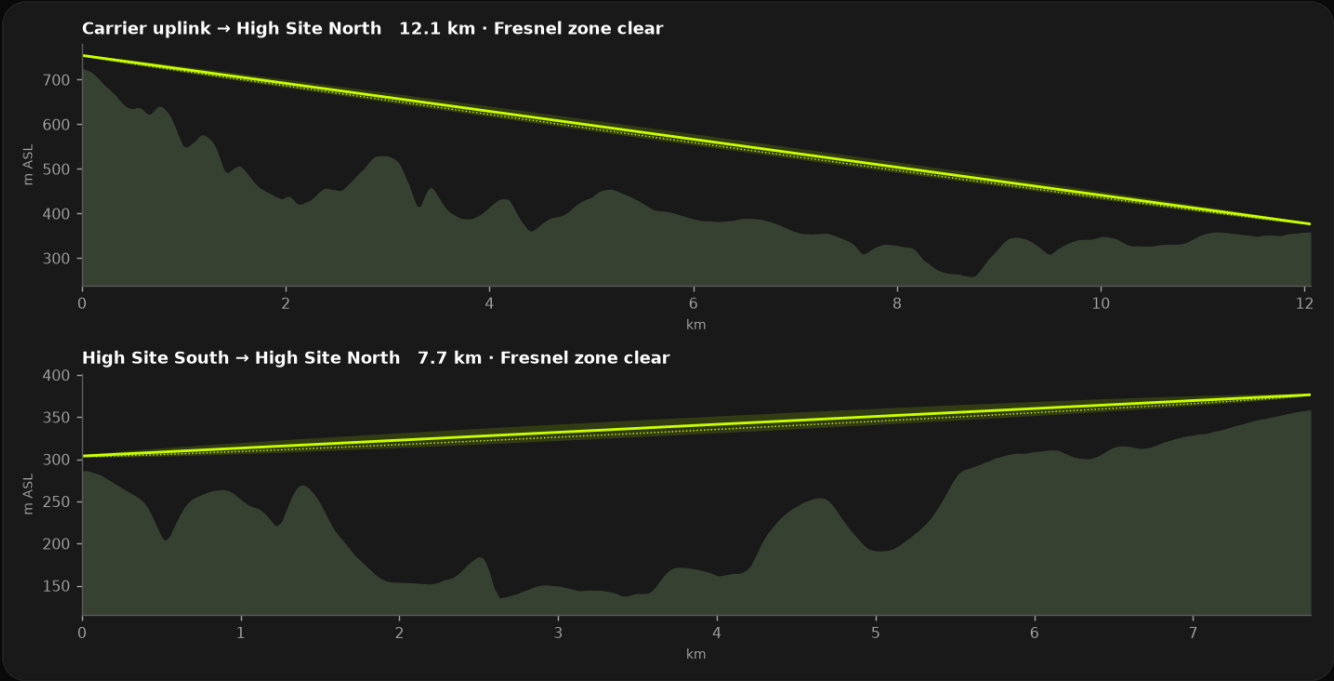
BUILT FOR WHAT COMES NEXT

Once the backbone covers the reserve, a new camera, sensor or lodge is an endpoint on a network that is already there. It is not a new infrastructure project or a new provider contract.

03

Every path checked against the terrain

Each profile shows the ground between two sites, the radio path (lime line) and its Fresnel zone (lime band). We allow for earth curvature and require at least 60% of the zone to be clear at every point. A path that fails is not drawn.



LODGE	SERVED FROM	PATH LENGTH	LINE OF SIGHT
Spekboom Bush Camp	—	—	POSITION TO CONFIRM
Acacia Lodge	—	—	POSITION TO CONFIRM
Khatu Villa	—	—	POSITION TO CONFIRM

Entries marked “position to confirm” have no published location. We add them to the model once you share a marked map or KMZ.

5.8 GHz · $k = 4/3$ · uplink tower 30 m · high-site masts 18 m · lodge masts 8 m. Lodge positions come from published GPS coordinates.

One backbone for every driver of the business

A reserve network carries three loads at once. This design is built to hold all three on one backbone.

GUEST EXPERIENCE

Reliable Wi-Fi, booking and card payments at every lodge.

STAFF COMMUNICATION

Lodges, HQ, rangers and maintenance on one network, day and night.

SECURITY AND OPERATIONS

Cameras, gates, fence alarms and sensors on the same backbone, powered through load-shedding.

What other properties got back

Sandymount Safaris 30,000+ ha, three lodges

BEFORE

A consumer wireless service at about 3 Mbps down and 8 Mbps up, with the provider on site daily to swap IPs and reinstall equipment.

AFTER

A Vodacom-backed 100 Mbps symmetrical link into the reserve, a CTTX-built main mast and two secondary masts, solar and lithium power, all monitored remotely.

GUEST EXPERIENCE

HD sport streams live at the lodges. Guest connectivity is no longer something the lodges think about.

STAFF COMMUNICATION

Over 800 Mbps measured across the reserve's own backbone, linking the main lodge, Mountain Retreat and Samar.

SECURITY AND OPERATIONS

Daily provider callouts went to zero. Every fault used to need a vehicle and a person; now the network is monitored remotely.

OWNERSHIP

The reserve keeps adding systems on top of the backbone. In the client's words, they stopped doing speed tests.

WHEN THE NETWORK GOES QUIET AFTER HOURS

Elephants broke out after 21:00 and staff had no effective way to coordinate. The requirement that came out of it: dependable communications after hours, across the areas where an incident can happen.

An Eastern Cape lodge near Addo

OWNING THE NETWORK PAYS FOR ITSELF

Replacing a rented, contended service and rented Wi-Fi with a lodge-owned network breaks even in under three years on connectivity savings alone. It also connects areas the old setup never reached, including the spa.

The same lodge, business case from its own invoices (Sep 2026)

06

What it pays back

A reserve that replaces a separate rented connection and rented Wi-Fi at every lodge with one owned backbone typically recovers the investment in under three years on connectivity costs alone. The security, operations and guest-experience benefits come on top of that.

YOUR NUMBER, NOT OURS

Send us each lodge's latest connectivity invoice, or just the monthly totals, and we will work out Buffalo Kloof's actual payback. It takes us a day.

BUILT TO BE FORWARDED

If infrastructure decisions sit with someone else, please pass this on. Owners will find the payback and ownership case, the reserve manager the security and operations case, and lodge managers the guest experience case.

What changes when the reserve owns it

TODAY

3

WITH A RESERVE BACKBONE

- One uncontended carrier feed shared across the reserve
- Lodges, HQ, the anti-poaching base and gates on one private network
- Coverage out into the field, day and night
- An owned asset that takes cameras and sensors without new towers

Avoided loss

Surveillance cameras, helicopter patrols and tracking already run on Buffalo Kloof. An owned backbone carries that data and ranger communications on one monitored network, day and night.

Operational efficiency

Water points, pumps and fuel tanks report their own status, so there are fewer vehicle trips just to check. Maintenance, ecology and lodges share one communications path.

Infrastructure ownership

High sites, backbone and power belong to the reserve. The network grows with the reserve instead of depending on each lodge's separate provider.

Executive risk reduction

Management gets live visibility of the network, cameras and sensors, and a record of operational diligence for owners, insurers and conservation partners.

Next steps

1

30-minute call

Your lodges' current connectivity costs, your peak usage, the incidents that were hardest to coordinate, and what currently needs a vehicle trip to check.

2

Confirm the sites

A marked map of HQ, the anti-poaching base and the gates. We add them to the model and rerun the link plan.

3

One day on the reserve

We walk the high sites and relay points, and confirm the carrier uplink with Vodacom.

4

Design and proposal

A final design in Cambium LINKPlanner, with equipment, power, a phased plan and the investment case.

Method. Terrain: Copernicus DEM GLO-30 (30 m). Imagery: ESA Sentinel-2 L2A true colour, 26 March 2026 to 14 July 2026. Radio model: 5.8 GHz, effective earth radius factor 4/3, 60% first-Fresnel-zone clearance at every point. The terrain model does not include trees or buildings. Every path is confirmed on survey before anything is built.

Candidate positions. High Site North -33.38606, 26.62901; High Site South -33.44867, 26.66586; carrier uplink -33.33075, 26.51738 (Makhanda high ground). The exact carrier site is subject to Vodacom feasibility.

Gerhard Smit

CTTX Services (Pty) Ltd

Tel 084 550 3281

Email gerhard@cttx.co.za

Web cttx.co.za